Trustworthy Borehole Database Ingestion from VLM Proposals: Provenance and Spatial Support

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ABSTRACT

Background: High visual extraction accuracy does not by itself establish a trustworthy geological database. A database row also needs independently checkable page evidence, a decision state, and an account of what abstention removes from downstream spatial support. **Methods:** We evaluate the frozen `Qwen/Qwen3.8-27B-FP8` direct page-to-JSON reader on five record-disjoint California cohorts (450 reports; 8,268 published manual-transcription intervals) and source-shift panels. The headline metric is boundary-pair interval F1: both interval depths must match under an order-preserving tolerance. We then add an independently positioned reader, deterministic depth/column checks, and an accept-or-review policy; a separate legacy sequence-reconstruction analysis is reported only as a harm analysis. **Results:** Qwen reaches boundary-pair F1 0.896–0.932 on California, but falls to 0.577 on the Swissgeol source-agreement panel. On held-out California v003, independent evidence yields accepted-interval precision 0.993 (444/447 accepted intervals correct) at 0.244 proposal coverage. Only 4/100 documents satisfy complete-document auto-acceptance, which defines a conservative deployment boundary rather than a claim of full automation. A spatial diagnostic shows that full-support risk-aware volume discrepancy is 0.0821 versus 0.1387 for raw extraction, while retaining only 0.636 of the reference convex-hull area; on the identical 15-document accepted subset, risk and rereading are both 0.0754 versus 0.0326 for raw. **Conclusions:** Modern VLMs are strong proposal readers, not database authorities. Provenance-grounded selective decisions must report precision together with coverage, complete-document utility, review burden, and the spatial-support consequences of abstention.

CRediT authorship contribution statement

Yifan Du: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization; Writing – original draft; Writing – review and editing.

Highlights

- Qwen3.8-27B-FP8 reaches 0.896–0.932 boundary-pair F1 on 450 reports.
- Independent evidence reaches 0.993 precision at 24.4% proposal coverage.
- Only 4% of held-out documents qualify for complete automatic acceptance.
- Abstention changes spatial support and can reverse apparent improvement.

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1. Introduction

Digitizing historical borehole logs is often described as an OCR problem. That description is operationally incomplete. A typical log combines a depth scale, cumulative boundary or thickness columns, sampling marks, lithology symbols, free-text descriptions, water-level annotations, and report metadata. The same number may be a top boundary, a bottom boundary, a sample depth, or a scale tick depending on its column and vertical context. A parser can therefore achieve a favourable character or interval score while creating an unusable geological record.

The central problem addressed here is the gap between visual extraction and trustworthy database acceptance. A trustworthy record must satisfy four conditions. First, the predicted interval sequence must be numerically coherent and sufficiently complete. Second, each critical value must retain provenance to a source page and region. Third, the system must distinguish a plausible proposal from an automatically acceptable update. Fourth, abstention must be assessed as a change in downstream support rather than treated as a harmless missing value.

Modern VLMs make the first condition substantially easier on familiar page families. They can associate typography, lines, symbols, and prose without a separate OCR transcript. That capability changes, rather than eliminates, the scientific question. If a direct VLM produces high boundary-pair F1 but the tested interface does not provide independently verifiable field provenance, a calibrated acceptance state, or an independent column-ownership witness, a database curator cannot tell which rows are safe to ingest. Conversely, a conservative system may reduce false corrections by rejecting difficult pages while deleting the spatial observations needed for a surface or volume diagnostic. We therefore evaluate extraction, assurance, and downstream consequence as one chain.

This paper tests one central hypothesis: high extraction capability becomes trustworthy database ingestion only when each automatically admitted interval has independently checkable provenance and deterministic structural checks, and when abstention is evaluated as a change in downstream spatial support. The hypothesis links capability, evidence, decision, and geoscientific consequence rather than treating them as separate benchmarks.

We retain exactly three research questions:

RQ1 — Extraction reliability under cohort and source shift. How stable are boundary-pair interval predictions and complete boundary sequences across independent cohorts from the same publication program and across unrelated source families?

RQ2 — Provenance-grounded selective assurance. Can independent positioned evidence, field semantics, deterministic depth geometry, and structural sequence constraints convert high-recall VLM proposals into auditable selective decisions with low critical error and explicit abstention?

RQ3 — Downstream consequence of extraction error and abstention. How do value errors, missing or altered ordered boundaries, and risk-driven support deletion propagate to spatial support, interpolated surfaces, and volume diagnostics?

The contributions correspond directly to the three questions:

1. **Capability and transport.** A multi-cohort and source-shift evaluation shows that modern VLM boundary extraction can be strong on a familiar borehole-log family while remaining insufficient for transportable, database-level assurance.

2. **Provenance-grounded selective assurance.** A VLM is treated as a proposal reader; independently positioned evidence and deterministic structural checks control which proposals can be automatically accepted. The legacy recovery analysis is a supporting harm experiment, not a separate method claim.

3. **Downstream support-aware evaluation.** Full-support and matched-support diagnostics show that abstention changes the geoscientific observation set and can change apparent downstream error. Reproducibility records provide the infrastructure for these claims rather than a fourth contribution.

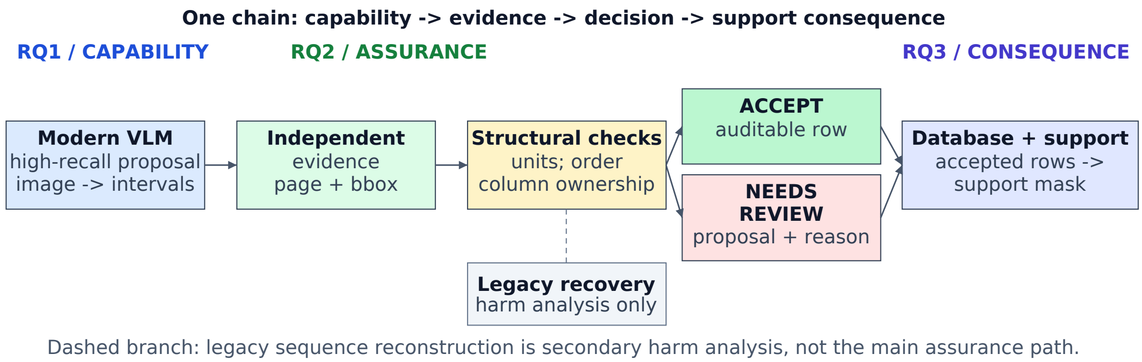

Figure 1: Provenance-grounded assurance framework. VLM proposals are checked against independent positioned evidence and deterministic geometry before acceptance or review.

2. Related Work

2.1. Borehole-log extraction and document intelligence

Document intelligence separates recognition, layout, table topology, and visual-language reasoning rather than treating them as one OCR capability (Smith, 2007; Xu et al., 2020; Pfitzmann et al., 2022; Smock et al., 2022; Kim et al., 2022; Hu et al., 2025). Borehole studies report image-based extraction (Zhang et al., 2020), OCR-assisted classification and database construction (Han and Suh, 2024), PDF workflow evaluation (Amini et al., 2023), historical well-record extraction with a large language model (Ma et al., 2024), and VLM-based structured generation (Shiga, 2026). These studies establish what a page reader can recover, but their targets and source families differ; extraction quality is therefore not automatically a database admission decision.

2.2. Selective prediction, provenance, and trustworthy ingestion

The reject option has a long statistical history: a classifier may reduce error by declining cases whose expected cost is too high (Chow, 1970). Modern selective-classification work formalizes coverage and selective risk (Geifman and El-Yaniv, 2017, 2019), while a recent survey organizes reject-option design and evaluation across machine-learning settings (Hendrickx et al., 2024). Conformal risk control provides a related way to state finite-sample risk targets, although the present protocol is a fixed evidence gate rather than a conformal procedure (Angelopoulos et al., 2024). Human-in-the-loop systems emphasize preserving the hand-off between model proposal and human decision (Amershi et al., 2014). Database provenance formalizes why a value exists and which source derivation supports it (Buneman et al., 2001; Simmhan et al., 2005). Together these works explain why prediction accuracy and acceptance authority are different objects.

2.3. Geoscientific uncertainty and spatial support

Geoscientific studies show that borehole interpretation and geological interfaces inherit uncertainty from observations and modelling choices (Lark et al., 2014; Pakyuz-Charrier et al., 2018; Wellmann and Caumon, 2018; Wang, 2023). Lithology vocabularies and automated stratigraphic metrics help compare structured interpretations, but they presuppose that intervals have already been recovered and semantically assigned (McCormick and Heaven, 2023; Fuentes et al., 2020; Garzón et al., 2026). Recent density studies further show that changing borehole support can alter geostatistical and three-dimensional model behaviour (Tran et al., 2025; Zhang et al., 2026).

Table 1: Representative-work comparison across extraction capability, provenance, selective admission, document-level utility, and downstream spatial support. “Not reported” means that the cited study did not report that evaluation dimension; it does not imply a methodological deficiency for the study’s stated objective.

Representative study	Domain/source	OCR/VLM	Cross-source	Provenance	Selective gate	Complete document	Spatial support
Zhang et al. (2020)	Borehole-log images	OCR/deep model	Not reported	Not reported	Not reported	Not reported	Not reported
Han and Suh (2024)	Abandoned-mine logs	OCR/deep model	Not reported	Page extraction, not field trace	Not reported	Structured target, not admission	Not reported
Amini et al. (2023)	Government PDF logs	OCR/PDF workflow	Multiple PDF sources	PDF selection and capture	Not reported	Not reported	Not reported
Ma et al. (2024)	Historical well records	Large language model	Not reported	Record extraction, not bbox gate	Not reported	Not reported	Not reported
Geifman and El-Yaniv (2017)	General classification	Neural classifier	Not applicable	Model confidence	Reject option	Coverage/risk, not documents	Not reported
Hendrickx et al. (2024)	Reject-option survey	Multiple ML settings	Review of settings	Not a source-provenance workflow	Selective risk	Not document completeness	Not reported

Table 1: Representative-work comparison (continued)

Representative study	Domain/source	OCR/VLM	Cross-source	Provenance	Selective gate	Complete document	Spatial support
Buneman et al. (2001)	Database provenance	Not an extraction model	Not applicable	Provenance semantics	Not an acceptance experiment	Not reported	Not reported
Lark et al. (2014)	Geological cross-sections	Geoscientific modelling	Borehole uncertainty	Input uncertainty	Not an extraction gate	Not reported	Support/model uncertainty
This study	Borehole pages to database	VLM + positioned evidence	California cohorts + shift	Page-local bboxes	Fixed ACCEPT/ NEEDS_ REVIEW	4/100 documents	Full/matched

Existing studies establish extraction capability, selective-risk principles, and downstream geoscientific uncertainty largely as separate problems. This study connects them at the document-to-database decision boundary. Extraction studies mainly ask whether a page can be parsed; selective prediction asks when a model should abstain; geoscientific uncertainty studies ask how imperfect observations affect downstream models. The present question is when a visually plausible borehole extraction can be admitted to a geological database, and what observational support is lost when it is not admitted. Table 1 summarizes the representative-work comparison.

3. Data, Evidence Tiers, and Task Definition

3.1. Evidence tiers

The evaluation separates evidence types before any metric is calculated; Table 2 defines the tiers used here.

Table 2: Evidence tiers and the claims each tier supports in this study. Evidence tiers are kept separate rather than pooled into a single accuracy estimate.

Evidence type	Meaning	Claims supported here
Published manual-transcription Gold	External institution transcribed source images and applied reported QC	Formal boundary-pair and semantic extraction accuracy within that release
Source-agreement reference	An explicit page table agrees with an authoritative database sequence	Source-specific transfer and downstream consistency, not representative human image GT

Table 2: Evidence tiers and supported claims (continued)

Evidence type	Meaning	Claims supported here
Authoritative metadata	Official IDs, coordinates, collars, or final-depth fields	Field agreement and spatial-support diagnostics
Machine Silver	Multiple machines or deterministic rules construct a reference	Agreement and development analysis only
Audit/no reference	No independent target	Coverage, runtime, schema validity, and failure mechanisms

Synthetic records are a separate controlled class with programmatically known labels. Evidence tiers are never pooled into a single accuracy estimate. In particular, source-agreement panels are not described as newly annotated human Gold, and synthetic dual-reader experiments are not described as two observed readers.

3.2. Primary California cohorts

The primary accuracy evidence is the USGS California lithology release and its paired public report links (Haugen et al., 2025; Borkovich et al., 2025). The publisher's metadata describes manual transcription from well-completion-report images, preservation of reported wording, and checks for sequencing, gaps, and final-depth completeness. The project formed mutually record-disjoint freezes v001–v005. The formal evaluation contains 450 reports and 8,268 intervals: v001 contributes 50 test reports and 697 intervals, while v002–v005 contribute 100 reports each and 1,770, 1,788, 1,944, and 2,069 intervals.

The selection flow begins with 12,732 deterministically paired reports and 225,150 exact-deduplicated intervals. Filters retain reports with 5–60 intervals, empty source comments, and adjacent continuity at least 0.99. County-diverse acquisition and record-disjoint freezing define the five cohorts. The resulting data are useful for independent replication but are not a random sample of every California log; the moderate interval-count and continuity filters are reported as part of the estimand.

3.3. Source-shift panels

The Swissgeol Thurgau held-out panel contains 35 source-agreement documents and 80 explicit top/bottom intervals. It is selected from pages whose published table agrees with the official database and is therefore a transfer panel rather than a national Gold benchmark. BGS Offshore contains 26 historical source groups and 341 graphic-log intervals joined from official survey, scan, and geology layers. Raft River contributes two reports with 62 explicit interval rows. These panels stress different page lengths, typography, scan quality, and column semantics. A one-time unseen BGS source-family run is reserved for the external risk gate and never used for tuning.

3.4. Structured spatial support

The downstream analysis uses the 35 Swissgeol documents and their authoritative coordinates/collars after extraction decisions are frozen. The 35 documents contain 80 ordered boundaries with sparse support at depth. A separate controlled support-preservation protocol is retained in Supplementary Methods S5; it is not part of the image-extraction benchmark and does not provide an observed second reader.

3.5. Task output and headline metric

The output schema stores borehole metadata, ordered intervals, raw and normalized terminology, page and bbox provenance, extraction method, confidence, validation state, and warnings. The headline interval metric is **boundary-pair interval F1**. Given reference intervals (G) and predictions (P), a prediction matches only when both top and bottom depths fall within the inclusive tolerance after unit conversion, under order-preserving maximum-cardinality matching with minimum total error as the tie-break. A single correct top boundary or a syntactically valid JSON object is not an interval match.

Boundary-exact rate requires the complete ordered boundary sequence for a document. Full-record exact additionally requires the evaluated semantic fields to agree. Matched lithology exact is reported when the reference and output contain comparable lithology fields. The direct Qwen evaluation schema and archived metrics do not include a validated lithology/full-record adjudication for every proposal; we therefore report boundary-pair F1 and boundary-exact for that model and explicitly mark semantic/full-record correctness as unavailable rather than infer it from JSON validity. The positioned OCR baselines provide matched-lithology and full-record exactness as a complementary lower-level diagnostic.

4. Provenance-Grounded Selective Assurance

4.1. Modern VLM proposal reader

The open modern baseline is the official `Qwen/Qwen3.8-27B-FP8` checkpoint (Qwen Team, 2026). The weights use fine-grained dynamic FP8 E4M3 and were served with vLLM over four RTX 2080 Ti GPUs. The serving process is **partially reconstructable**: its source commit, per-request logs, and immutable contemporaneous runtime lockfile were not recoverable. The full component hashes, local revision, runtime versions, and unrecoverable fields are retained in Supplementary Table S4; this main-text summary identifies the evaluated object without interrupting the scientific narrative.

Pages are rendered at 200 DPI with PyMuPDF to lossless PNG, without crop, rotation, or enhancement. The prompt is `vlm_interval_source_units_v002`, SHA-256 `891bc6beb7ff9cf35c55389191a208c9b09e9e2dc76909f716603f41374 5104a`. Decoding uses temperature 0, provider-default top-p, thinking disabled, a 4,096-token maximum, zero automatic retries, and strict JSON parsing. No repair, reordering, deduplication, or reference-conditioned completion is performed. Non-finite or non-positive ranges are rejected during deterministic source-unit conversion. This protocol was frozen before each cohort result.

The VLM returns interval proposals $V = (v_1, \ldots, v_n)$. It is intentionally treated as a reader that proposes visible structure, not as an authority that can directly write a database. This separation allows the paper to evaluate the value of modern visual semantics without granting the model unobserved provenance or acceptance authority.

4.2. Independent positioned evidence

An independently frozen RapidOCR positioned parser produces candidates $C = (c_1, \ldots, c_m)$. Each candidate retains page index, normalized top and bottom column positions, y-order, OCR confidence, geological-term evidence, source text, and the original region bbox. The VLM and positioned parser do not exchange outputs. A proposal and positioned candidate are eligible for evidence agreement only when their top and bottom depths agree under a strict source-unit tolerance and the candidate retains both source regions.

The numeric anchor is weaker than semantic ownership: finding the same number somewhere on a page does not prove that it is the interval boundary. Automatic acceptance therefore requires complete top-bottom interval agreement, monotone order, non-overlap, positive thickness, and retained bboxes. Partial agreement is recorded as a proposal for review, not as a complete accepted interval.

4.3. Deterministic geometry and evidence gate

The positioned reader supplies candidates with source-unit top and bottom depths, page order, normalized column coordinates, OCR confidence, geological-term evidence, source text, and both original region bboxes. Deterministic checks convert these fields into an evidence state: finite and positive depths, compatible units, top < bottom, non-overlap, monotone page/depth order, and agreement of both endpoints with the same positioned semantic column. A matched number without semantic ownership remains `LOCATED`, not `ACCEPTED`.

The main assurance path is therefore a four-stage decision, shown in Fig. 1: (i) the VLM proposes visible intervals; (ii) the independently positioned reader supplies page-local evidence; (iii) deterministic geometry checks test endpoint, unit, order, and ownership conditions; and (iv) the system returns `ACCEPT` or `NEEDS_REVIEW` while preserving the immutable proposal and its provenance. The direct VLM interface used here did not provide independently verifiable field provenance; the assurance layer adds that missing evidence rather than assuming that JSON validity supplies it.

4.4. Selective accept/review policy

The policy accepts a proposal only when both endpoints agree with an independently positioned interval, both source bboxes are retained, the interval is non-overlapping and monotone, and no critical numerical or unit check fails. Partial agreement is recorded as a proposal for review. The protocol was frozen after California v001 development: the agreement and field-evidence tolerances are fixed at 10^{-6} in their respective units, v002 is validation, and v003 is the reported held-out replication. Although v004 and v005 are held-out direct-reader cohorts in the broader protocol, no VLM assurance confirmation run is claimed for them here. The document is the primary risk unit because actions cluster within reports; action-level false-correction rates are secondary.

The policy reports three different quantities rather than one automation score: proposal coverage (accepted intervals divided by VLM proposals), selective precision among accepted intervals, and complete-document automation (documents for which the full ordered record passes the acceptance gate). On held-out v003, complete-document auto-acceptance is 4/100 (4%). This is the intended conservative deployment boundary: it identifies a small set that can enter a database automatically and sends the remaining proposals to review, rather than treating partial acceptance as complete-record correctness. For (n) accepted documents and zero observed worsened documents, the one-sided 95% upper bound is (1 - 0.05^{1/n}); this finite-sample statement is not a safety certification.

4.5. Secondary legacy sequence-reconstruction and harm analysis

The positioned candidate pool also supports a separate, legacy sequence-reconstruction analysis. It is not the executed end-to-end VLM assurance path. Its purpose is diagnostic: quantify how monotonic path selection, continuity, column stability, and semantic scores change recall and false corrections when proposals are reconstructed without the independent VLM-evidence gate. The full candidate representation, dynamic-programming objective, threshold grid, and same-pool ablation are specified in Supplementary Methods S2. This separation prevents a recovery-oriented decoder from being mistaken for the conservative acceptance policy.

4.6. Spatial consequence protocol

For boundary r in borehole i, elevation is $z_{ir} = c_i - d_{ir}$, where c_i is collar elevation and d_{ir} is depth. IDW at query location u is

$$\hat{z}_r(u) = \frac{\sum_{i \in N(u)} \|u - u_i\|^{-p} z_{ir}}{\sum_{i \in N(u)} \|u - u_i\|^{-p}}. \tag{1}$$

Thickness is the difference between adjacent surfaces. For a hull-clipped grid G, the volume diagnostic is

$$\hat{V}_\ell = A|G|^{-1} \sum_{u \in G} \hat{h}_\ell(u), \tag{2}$$

and the aggregate reference-relative volume discrepancy is

$$\frac{\sum_\ell |\hat{V}_\ell - V_\ell|}{\sum_\ell |V_\ell|}. \tag{3}$$

We report two estimands. Full-support comparison lets each extraction policy use its own available points; it measures the deployed package, including selection. Matched-subset comparison restricts raw, reread, and risk-aware inputs to the same 15 accepted documents; it measures value/sequence differences conditional on acceptance. If risk and reread are identical on this subset, any full-support difference is selection and support, not an additional correction. Spatial diagnostics include point coverage, convex-hull area ratio, nearest-neighbour distance, grid-to-nearest-observation distance, IDW power/neighbour/grid sensitivity, leave-one-borehole-out error, and volume jackknife. Abstention is treated as a **spatial sampling operator**: it changes the set of observations available to the downstream diagnostic, not merely the values attached to fixed locations.

5. Experimental Design and Reproducibility

All California cohorts are project- and record-disjoint. The direct Qwen protocol is fixed before evaluation, and no Gold error case changes prompt, decoder, matcher, threshold, or model roster. Source-shift panels are scored with their declared evidence tier. Swissgeol development and held-out roles are separated by salted PDF-content groups. The BGS v003 external run is opened once after the development gate and its result is not used for tuning.

The evaluation protocol and deterministic post-processing components are reproducible given the frozen page renders, candidate pools, configuration files, and seeds; bitwise deterministic VLM execution is not claimed. Document-cluster bootstrap resamples whole reports. Public release metadata records source URLs, exact manifests, hashes, evidence tiers, and archive checksums. The article package releases structured/reanalysis assets, manifests, hashes, source URLs, and recomputation materials. The separately versioned `data-v002` companion contains the selected source files and structured datasets that passed the author's item-level rights, attribution, linkage, privacy, sensitive-location, and embedded-content review. Model weights and private credentials are not redistributed.

6. Results

6.1. Capability is high on familiar sources but does not transport uniformly

On the five California cohorts, Qwen3.8-27B-FP8 obtains boundary-pair interval F1 of 0.932, 0.896, 0.918, 0.917, and 0.903. Its document-cluster 95% intervals are [0.888, 0.973], [0.841, 0.943], [0.878, 0.953], [0.876, 0.952], and [0.864, 0.939]. Complete boundary sequences occur in 74%, 70%, 72%, 74%, and 69% of documents. Zero-output rates are 0, 0, 0, 0.05, and 0.01. The corresponding frozen RapidOCR parser obtains 0.390, 0.450, 0.383, 0.428, and 0.389, with zero-output rates of 0.08–0.23 and full-record exactness of 0–2% in the available evaluated fields. The paired document-cluster F1 gains of Qwen over RapidOCR are 0.542, 0.445, 0.535, 0.489, and 0.514; every bootstrap probability that the gain is positive is 1.000. Figure 2 and Table 3 report the same F1 results.

The direct VLM is therefore not a failure case. It recovers visually implicit structure that the positioned parser misses. However, JSON validity is not provenance. Before deterministic rejection, invalid numeric ranges occur at 0.004–0.017 across the five cohorts. The tested direct interface did not provide independently verifiable field provenance, an independent column-ownership witness, or an auditable acceptance trace. The headline F1 is a boundary-pair metric; it does not establish lithology correctness or complete semantic records. In the RapidOCR reference, matched lithology exactness ranges from 0.431 to 0.544 and full-record exactness remains 0–2%. The source-shift panels make the deployment implication concrete: Swissgeol Qwen F1 is 0.577 with zero boundary-exact documents, while BGS Offshore RapidOCR and Tesseract obtain 0.038 and 0.041. On the one-time unseen BGS external gate, all five visible pages are classified as unsupported and abstain, producing zero utility but no false positive or critical numerical error. These results are tiered source-shift evidence, not a pooled benchmark; they show that capability does not establish transportability or database assurance.

An exploratory open-model roster tests whether this transport risk is specific to the 27B serving path. On a fixed 20-page California v003 panel, Qwen3-VL-4B-Instruct reaches boundary-pair F1 0.793 (207/249 matched intervals; 3/13 complete documents); on the complete 35-page Swissgeol panel it reaches 0.619 (43/80; 0/35 complete documents). PaddleOCR-VL-1.6 and MinerU2.5-Pro complete page inference on Swissgeol but produce no rows through the fixed auditable interval decoder. Those specialist results are reported as task/interface coverage, not visual recognition failure. The bounded interpretation is recurring transport or usable-output risk across model families and interfaces, not a universal capability estimate (Supplement S4.1).

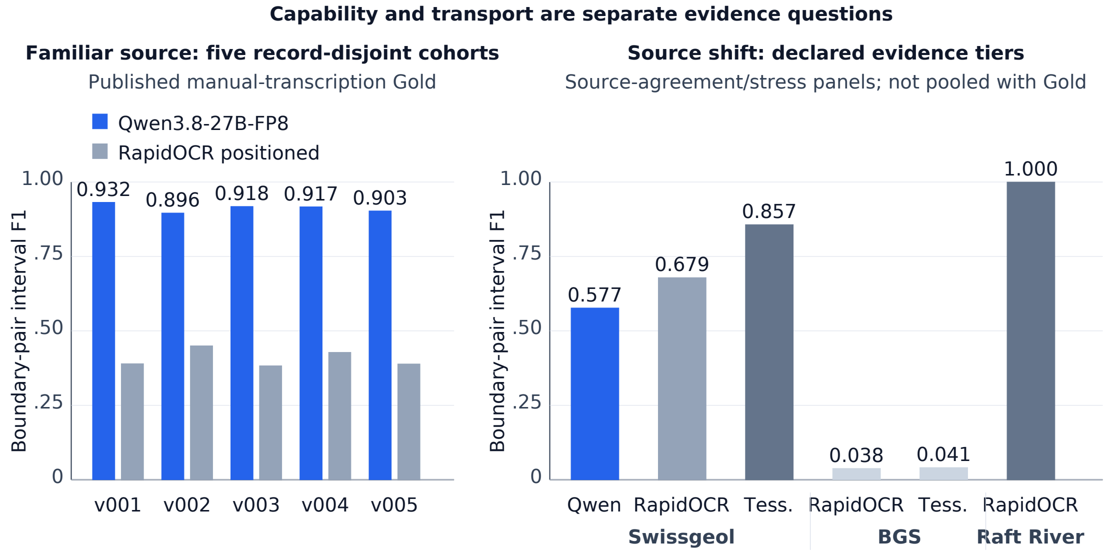

Figure 2: Modern VLM reliability across California cohorts and source shift. The familiar-source panel compares Qwen3.8-27B-FP8 with the positioned RapidOCR parser on five record-disjoint California cohorts with published manual-transcription Gold evidence. The source-shift panel reports Swissgeol, British Geological Survey (BGS), and Raft River transfer/stress outcomes with their declared evidence tiers; these values are not pooled with California Gold.

Table 3: Boundary-pair interval F1 across cohorts, readers, and source-shift panels. These are the results shown in Fig. 2; evidence tiers are declared separately and are not pooled.

Panel	Reader/interface	Evidence tier	Documents	Reference intervals	Boundary-pair interval F1
California v001	Qwen direct	Published manual-transcription Gold	50	697	0.932
California v001	RapidOCR positioned	Published manual-transcription Gold	50	697	0.390
California v002	Qwen direct	Published manual-transcription Gold	100	1,770	0.896
California v002	RapidOCR positioned	Published manual-transcription Gold	100	1,770	0.450
California v003	Qwen direct	Published manual-transcription Gold	100	1,788	0.918
California v003	RapidOCR positioned	Published manual-transcription Gold	100	1,788	0.383
California v004	Qwen direct	Published manual-transcription Gold	100	1,944	0.917
California v004	RapidOCR positioned	Published manual-transcription Gold	100	1,944	0.428
California v005	Qwen direct	Published manual-transcription Gold	100	2,069	0.903
California v005	RapidOCR positioned	Published manual-transcription Gold	100	2,069	0.389
Swissgeol held-out	Qwen direct	Source-agreement reference	35	80	0.577
Swissgeol held-out	RapidOCR positioned	Source-agreement reference	35	80	0.679
Swissgeol held-out	Tesseract positioned	Source-agreement reference	35	80	0.857
BGS Offshore	RapidOCR positioned	Source-agreement reference	26	341	0.038
BGS Offshore	Tesseract positioned	Source-agreement reference	26	341	0.041
Raft River	RapidOCR positioned	Source-agreement reference	2	62	1.000

6.2. Independent evidence creates a high-precision but selective operating point

The assurance experiment keeps the Qwen proposals unchanged and adds an independently positioned reader. On development v001, complete positioned agreement covers 174/736 proposals and all are correct. On validation v002, 561 proposals are accepted at precision 0.979 [0.951, 0.997], compared with raw proposal precision 0.854. On held-out v003, the frozen rule accepts 447/1,833 proposals: 444 are correct, selective precision is 0.993 [0.984, 1.000], raw proposal precision is 0.907, and accepted coverage is 0.244. The three incorrect accepted intervals occur in three different documents. Same-page numeric-anchor coverage is an **endpoint-field** quantity: 3,099/3,666 proposed top/bottom fields are located (0.845). Requiring both endpoints to be anchored yields 1,450/1,833 proposals (0.791), still substantially higher than the 447/1,833 semantically owned and accepted intervals. Finding a number is therefore easier than proving interval ownership. Figure 3 visualizes the operating point and evidence funnel; Table 4 reports the corresponding values.

The selective result is intentionally not reported as whole-document accuracy. Partial proposal acceptance cannot establish complete-record correctness, and non-accepted proposals remain in the review queue. Complete-document auto-acceptance is only 4/100 documents (4%) on held-out v003, even though interval-level accepted coverage is 24.4%. That gap is not a weakness hidden by the metric; it is the conservative deployment boundary. The method's benefit is a reliable subset with explicit provenance, not an assertion that the unaccepted 75.6% are correct. This distinction is operationally important when a database ingestion job must state which rows were automatically accepted and which require review.

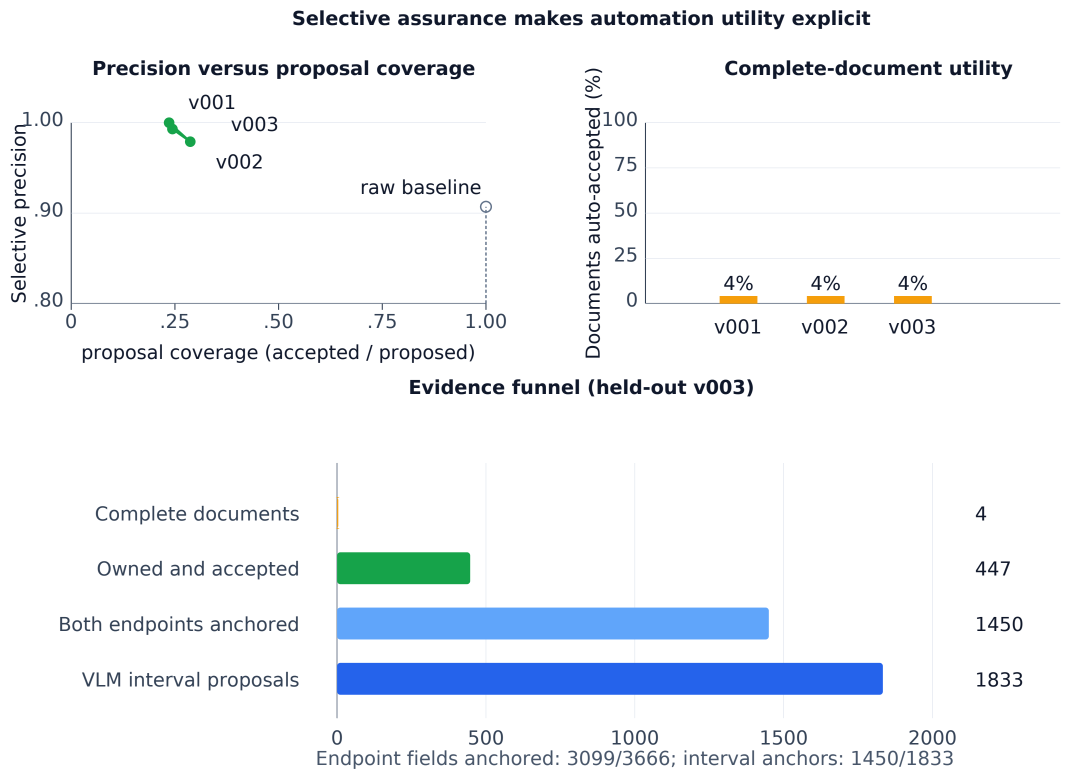

Figure 3: Selective assurance operating point. Precision, proposal coverage, complete-document automation, and held-out v003 evidence are shown together. Endpoint-field anchors are 3,099/3,666, both-endpoint interval anchors are 1,450/1,833, and semantically owned accepted intervals are 447/1,833; endpoint-field coverage and interval-level coverage use different denominators. The raw point is an unselective proposal baseline, not another selective cohort operating point.

Table 4: Independent evidence and selective assurance. Raw proposal precision, endpoint-field anchor coverage, both-endpoint interval coverage, and accepted coverage use distinct quantities and denominators.

Cohort	Raw proposal precision	Endpoint-field anchor coverage	Both-endpoint anchor coverage	Accepted coverage	Accepted interval count	Selective precision (95% CI)	Error docs
California v001 development	0.908	0.817	0.731	0.236	174	1.000 [1.000, 1.000]	0
California v002 validation	0.854	0.849	0.792	0.287	561	0.979 [0.951, 0.997]	5
California v003 held-out	0.907	0.845	0.791	0.244	447	0.993 [0.984, 1.000]	3

6.3. Unselective recovery illustrates why accuracy gains are not sufficient

On the same candidate pools, unselective sequence ranking increases California interval F1 from 0.383–0.450 to 0.470–0.566. The gain is driven primarily by monotonic path selection: on v004/v005, monotonic decoding reaches F1 0.579/0.550, while adding continuity, column stability, and the semantic bonus moves the operating point toward precision without a consistent F1 gain. This is a secondary mechanism analysis, not the main assurance path.

The recovery gain is not automatically safe. Unselective reconstruction produces action-level FCR 0.084–0.210 across the five cohorts and lowers document F1 on multiple reports. The addition-only policy accepts 43 additions on v004 and 39 on v005, for 82 actions in 19 documents. No accepted action is incorrect and no document is observed to worsen, but the primary one-sided document-level 95% upper risk bound is 0.1459. The action-level iid bound is smaller and secondary because actions cluster within reports. The policy yields a net gain of 41 matched intervals per 100 documents, compared with 230.5 under unselective reconstruction. These results are secondary harm analysis, not evidence that the legacy decoder is the main VLM assurance algorithm: automatic acceptance is reliable on a small subset, while most potential recovery remains review or abstention.

6.4. Abstention changes downstream spatial support

On the 35-document source-agreement panel, full-support raw, reread, and risk-aware inputs produce reference-relative volume discrepancies 0.1387, 0.1213, and 0.0821, with mean thickness MAE 45.623, 45.350, and 34.899 m. A superficial reading would call risk-aware selection an improvement. It would be incomplete. Risk accepts 15/35 documents and retains only 0.636 of the reference convex-hull area for the first boundary. Mean nearest-neighbour distance rises from approximately 1.39 km to 3.48 km, and mean grid-to-nearest-observation distance rises from 2.75 km to 4.62 km. Accepted documents have raw aligned boundary MAE 0.000 m and 12/15 exact raw sequences; rejected documents have 19.550 m and 13/20 exact sequences. The router selects easier and spatially narrower records.

The matched-subset estimand removes this selection difference. On the identical 15 accepted documents, raw, reread, and risk-aware thickness MAE are 35.128, 34.670, and 34.670 m, while reference-relative volume discrepancy is 0.0326, 0.0754, and 0.0754. Reread and risk are identical after conditioning on acceptance. The lower full-support risk discrepancy therefore cannot be attributed to an additional correction; it is a consequence of selection and changed support. IDW sensitivity yields overlapping ranges, and reference-input leave-one-borehole-out MAE is 47.06 m across 80 targets. Full-support volume-jackknife means are 0.1348 [0.0884, 0.1659] for raw, 0.1177 [0.0489, 0.1699] for reread, and 0.0849 [0.0274, 0.1365] for risk-aware input. The overlap is more scientifically informative than the ordering of three full-data estimates. Figure 4 contrasts the estimands, and Table 5 reports the corresponding diagnostics.

The controlled perturbation mechanism study is retained in Supplementary Methods S5 because its channels are synthetic rather than observed readers. It is used only to illustrate why a deployed risk policy should track both value confidence and the support cost of abstention.

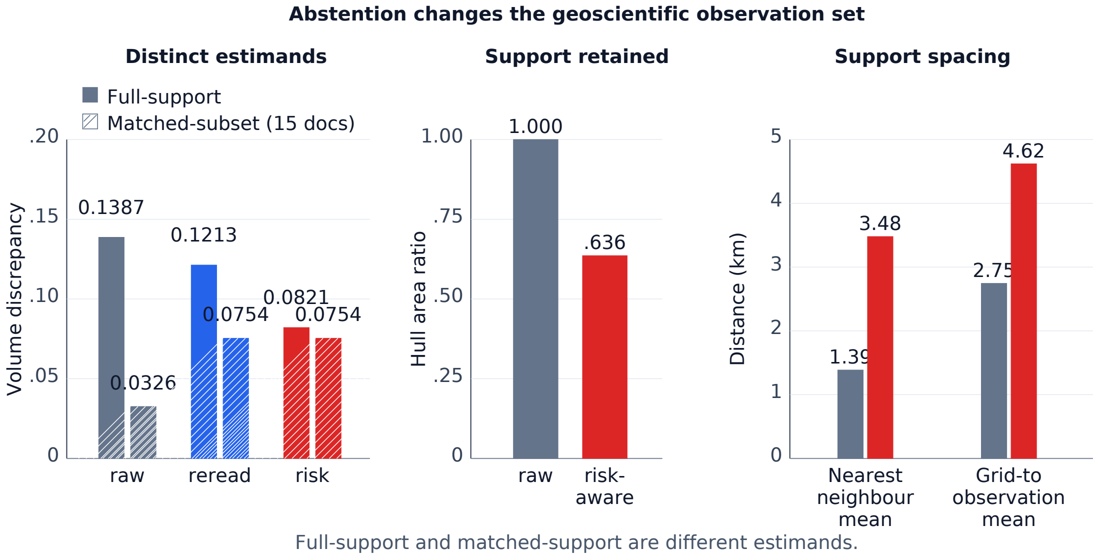

Figure 4: Full-support and matched-support downstream consequences. Solid bars use each extraction policy's available observations, whereas hatched bars use the identical 15 accepted documents. These are distinct estimands, not a direct value-correction comparison. NN denotes nearest-neighbour distance; grid distance denotes grid-to-nearest-observation distance.

Table 5: Risk, coverage, and downstream support diagnostics.

Analysis	Raw	Reread	Risk-aware
Full-support volume discrepancy	0.1387	0.1213	0.0821
Matched-subset volume discrepancy	0.0326	0.0754	0.0754
First-boundary hull-area ratio	1.000	1.000	0.636
Default LOO MAE (m)	49.84	46.62	47.05

7. Discussion

7.1. Capability versus assurance

The California result should be read as a capability result. Qwen recovers boundary pairs that the positioned OCR parser misses, and its 0.896–0.932 F1 range is a substantial improvement on this familiar publication family. It does not follow that the resulting rows are ready for ingestion. A boundary-pair metric does not test whether a number came from the depth column rather than a scale tick, whether a lithology phrase belongs to the same interval, whether a field can be traced to a source bbox, or whether a syntactically valid page contains a complete ordered record. The Swissgeol drop to 0.577 and the BGS zero-utility external gate show that source family and page semantics remain first-order variables. The appropriate deployment object is therefore not a score but a record with a provenance state and an explicit disposition.

7.2. Precision, coverage, and automation utility

Selective precision answers "how often is an accepted interval correct?" It does not answer "how much of the workload is automated?" On v003, 0.993 precision is achieved at 0.244 proposal coverage, while only 4/100 documents pass complete-document auto-acceptance. These are complementary facts. Reporting only precision would make a small accepted subset look like a complete database; reporting only coverage would penalize a policy for refusing unsupported structure. The three-layer report—selective precision, proposal coverage, and complete-document automation—makes the operating point visible and aligns the metric with a review workflow. The same logic explains why zero observed harmful actions are reported with a document-level finite-sample bound rather than a claim of zero risk.

7.3. Abstention as a geoscientific sampling decision

In a downstream geoscience workflow, rejecting a document does more than leave one value blank. It removes a coordinate, changes the convex hull, changes nearest-neighbour distances, and changes which grid cells are supported by observations. The full-support risk-aware discrepancy of 0.0821 is therefore not a pure correction effect. The matched 15-document comparison reverses the superficial ranking: risk and rereading are identical at 0.0754, while raw is 0.0326. This is the central computational-geoscience insight of the integrated study. An acceptance policy must be evaluated as a joint value-and-support operator, with full-support and matched-support estimands reported together. A lower error on a smaller, easier spatial subset is useful for triage but does not support claims of improvement for all geological models.

7.4. Practical implications

The practical deployment object is a record with four linked fields: proposal, independent evidence, decision, and support mask. A modern VLM supplies high-recall visual proposals; the positioned reader and deterministic checks control automatic acceptance; unsupported pages remain reviewable; and downstream analyses receive the accepted records together with the support mask. Full execution provenance is provided in Supplementary Table S4.

8. Limitations and Threats to Validity

First, the five California cohorts are independent but come from one publication program. They establish cohort stability, not national or multilingual generalization. Swissgeol and BGS are source-agreement or stress panels with different evidence tiers, and their results must not be pooled with California Gold. The BGS zero-utility external result is a transport diagnosis, not a claim that all BGS pages are unsupported.

Second, direct-VLM semantic correctness is incompletely observed. The Qwen interface was scored for boundary-pair interval F1, boundary-exactness, numeric invalidity, and transport. The archived direct-VLM protocol does not provide a validated lithology/full-record target for every proposal, so semantic correctness cannot be inferred from JSON validity or boundary matching. Future releases should add field-level semantic adjudication with the same evidence hierarchy.

Third, foundation-model contamination cannot be ruled out. California report identifiers, lithology strings, and public USGS tables may have appeared in pretraining data. Record-disjoint and source-disjoint splits reduce ordinary leakage, but they do not establish that the model has never seen the source family. The Qwen checkpoint, revision fields, component hashes, prompt hash, render settings, and partially reconstructable runtime record are therefore reported explicitly, and the results are interpreted as an operational benchmark rather than a contamination-free capability estimate.

Fourth, the provenance layer is conservative and incomplete by design. Independent positioned agreement is not statistical independence when both readers observe the same ambiguous page. A matched bbox proves localization, not geological truth. The 0.993 selective precision result has three errors and a document-level finite-sample bound; it is not safety certification. The direct closed host-managed visual pilot is excluded from headline comparisons because it contains only five pages and lacks provider trace, field bboxes, and complete runtime metadata.

Fifth, the spatial analysis is a diagnostic. Coordinates and collars are authoritative for the selected source-agreement panel, while page-derived coordinates are not fully available. IDW is transparent but not a complete geological modelling workflow. The supplementary controlled perturbation channels are synthetic and are not independent locations. Ordered boundary indices are not a substitute for geological unit correlation, faults, anisotropy, or uncertainty-aware 3-D modelling.

Finally, abstention changes the population on which a downstream model operates. Full-support and matched-subset results answer different questions and should both be reported in deployment. A system that improves surface error by discarding difficult, spatially important boreholes may be useful for a conservative review queue but should not be described as improving all geological models.

9. Conclusions

Modern VLMs materially improve borehole-log extraction on a familiar source family: `Qwen/Qwen3.8-27B-FP8` achieves 0.896–0.932 boundary-pair interval F1 and 69–74% complete boundary sequences across five California cohorts. That result is important, but it is not the same as a trustworthy geological database. The model's direct interface emits invalid numerical ranges, did not provide independently verifiable field provenance in this evaluation, and loses performance under source shift.

Provenance-grounded assurance addresses this gap without discarding modern visual semantics. An independently positioned reader, deterministic depth/column checks, and a selective accept/review policy convert a subset of high-recall VLM proposals into auditable decisions. Held-out California v003 reaches selective precision 0.993 at 0.244 coverage, but only 4% of documents are completely auto-accepted. The separate legacy sequence-reconstruction analysis demonstrates why F1 gains alone are unsafe; it is not conflated with the main assurance path.

The downstream analysis establishes the second half of the argument. Abstention is not a neutral null value: it deletes spatial support. Full-support risk-aware volume discrepancy appears lower, but matched-subset analysis shows that risk and reread are identical and both worse than raw for that selected subset; the apparent gain is selection and support geometry. Interpolation sensitivity and reference-input LOO error are of the same order as extraction-policy differences.

The resulting deployment principle is simple: use modern VLMs for high-recall visual proposals, require provenance-grounded independent evidence for automatic acceptance, preserve immutable raw outputs, abstain on unsupported structure, and report how acceptance changes downstream spatial support. High F1 is a useful capability indicator. It is not, by itself, a reliability argument for a geological database.

Computer Code Availability

The GeoLogParser repository contains versioned code/configuration, prompt hashes, metric bindings, figure generators, claim audits, and recomputation scripts for this article. The source code is released under the MIT license at `https://github.com/Entropic-Silence/GeoLogParser`. The complete article and result-reproduction package is identified by the annotated release tag `paper4-cageo-v1.0.1`. An archival DOI will be added after deposit.

Data Availability

The `paper4-cageo-v1.0.1` package contains the manuscript, supplement, figures, structured/reanalysis inputs, aggregate metrics, manifests, checksums, source URLs, and recomputation scripts needed to reproduce the reported result-level analyses. The separate `data-v002` companion contains the author-reviewed selected source files and structured datasets used by the principal experiments; it is a data companion, not the complete Paper 4 package. Source-specific terms and attribution remain in the release ledger, and linkable spatial inputs are not represented as anonymous. Model weights and private credentials are not redistributed. Archival DOI fields will be added after deposit.

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No claim in this manuscript relies on undisclosed human annotation, hidden reference-conditioned tuning, or a closed-model score that lacks a reproducible execution record.

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